9.3 States of Matter and Changes of State

4th Grade Science · Video: States of Matter and Changes of State (Smile and Learn)

Name: _____ Date: _____

Watch the video and fill in the blanks as you go.

Three States of Matter

Matter is the _____ all things are made of.

Matter exists in three states: liquid like juice, solid like a popsicle, and _____.

The water _____ coming out of a hot pot is a gas.

Solids, Liquids, and Gases

A liquid takes the _____ of its container. Its volume always stays the same.

In a liquid, the particles can _____ and slide past each other.

A solid has a fixed shape and _____.

In a solid, particles are held together in an _____ structure.

A gas can change its volume and shape. It takes up the space of its _____.

In a gas, particles are well _____ and move freely.

Heating Changes the State

A change in state happens when we heat or _____ matter.

If we heat a solid, it becomes a liquid. This is called _____, or melting.

If we heat a liquid, it becomes a gas. This is called _____. (Evaporation and boiling are kinds of vaporization.)

When a solid warms up, its particles start _____ from one another and move more freely.

Cooling Changes the State

If we cool a gas, it becomes a liquid. This is called _____.

Water vapor cools when it reaches the pot's _____ and turns back into a liquid.

If we freeze a liquid, it becomes a solid. This is called _____, or freezing.

When matter cools, its particles come _____ together and move more slowly.

Vocabulary

Word	Meaning
fusion (melting)	a solid changes to a liquid when it is heated
vaporization	a liquid changes to a gas when it is heated
condensation	a gas changes to a liquid when it is cooled
solidification (freezing)	a liquid changes to a solid when it is cooled
volume	how much space matter takes up

Key Takeaways

• Heating makes particles move faster and spread apart: solid to liquid to gas.

• Cooling makes particles slow down and come closer: gas to liquid to solid.